Take-Home Assignment: Vancouver Parcel Lot Analysis Tool

Time limit: 8 hours

Tech Stack

	Requirement
Framework	React 18+
Language	TypeScript (strict mode)
Build Tool	Vite
Map	Mapbox GL JS

Data Sources

- Parcel data: Vancouver Open Data — Property parcel polygons (GeoJSON)
- Road data: Mapbox Streets base map
- Mapbox Token: Use a free Mapbox account — the free tier is sufficient.

Lot Classification Rules

Edge Types

Type	Definition
Frontage	Edge adjacent to the parcel's primary street (matched by street name from parcel attributes + spatial proximity to road centerline)
Flankage	Edge adjacent to a secondary street (a different street from the primary). Typically seen on corner lots.
Rear Lane	Edge adjacent to a lane or alley
Rear	Edge on the opposite side of the Frontage that is **not** adjacent to any road or lane
Side	Any remaining edge connecting the Frontage/Flankage side to the Rear/Rear Lane side

Frontage vs. Flankage

Both Frontage and Flankage edges are adjacent to a street, but they are distinguished as follows:

- Frontage is the edge whose adjacent street name matches the parcel's address (e.g. if the parcel address is "123 Main St", the edge nearest to "Main St" is the Frontage).
- Flankage is any edge adjacent to a different street that does not match the parcel's address. This typically occurs on corner lots where the parcel borders two intersecting streets.
- In most cases, the Flankage edge (along the lot's depth) is longer than the Frontage edge (along the lot's width).

Lot Types

Lot Type	Condition
Corner Lot	Has both a Frontage edge and a Flankage edge (adjacent to two different streets)
Double Fronting	Has 2 Frontage edges facing different streets (opposite sides both face streets, not lanes)
Standard with Lane	1 Frontage edge + a Rear Lane edge on the opposite side
Standard without Lane	1 Frontage edge + a Rear edge on the opposite side (no adjacent lane)

Priority: Corner Lot > Double Fronting > Standard with Lane > Standard without Lane

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Requirements

1. Map & Parcel Interaction

- Mapbox map centered on downtown Vancouver with parcel polygons rendered.
- Clicking a parcel highlights it.

2. Address Search with Autocomplete

- Search bar with autocomplete dropdown, scoped to Vancouver only.
- Keyboard navigation (arrow keys, Enter, Esc) and input debounce.
- Selecting a result triggers the same flow as clicking a parcel (fly to, highlight, analyze, show info card).
- Show search history when the search bar is focused and the input is empty. A "Clear history" button should be visible when the history list is displayed.

3. Parcel Info Card & Classification

When a parcel is selected (via click or search):

- Info card showing: full address, area (m^2), primary street, and lot type label.
- Edge visualization on the map.

4. Debug Mode

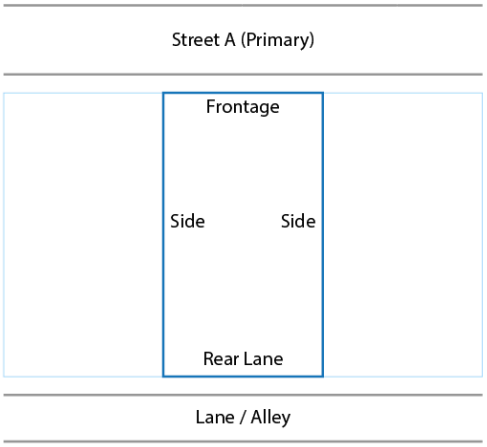
A toggle button that reveals developer diagnostics. You can decide which tools are needed in this mode.

Submission

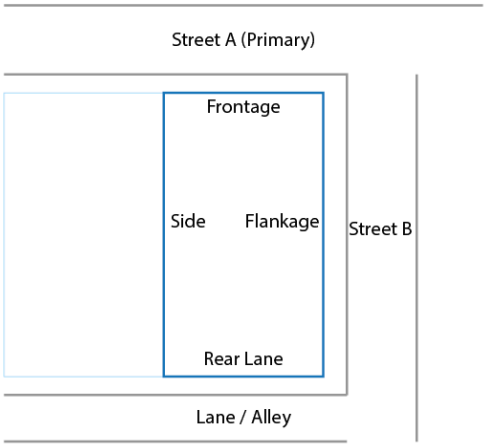
Upload to your own private GitHub repository. Present the repository during the review.

Appendix: Lot Type Diagrams

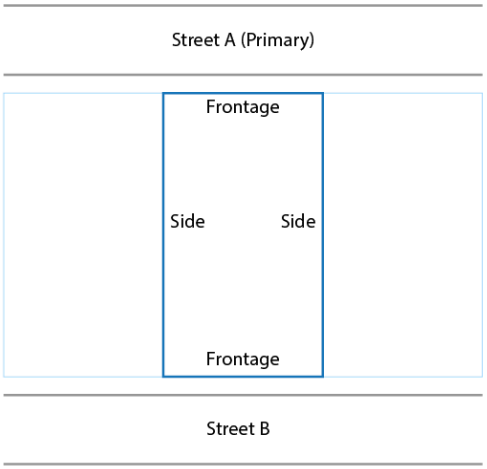
STANDARD WITH LANE



CORNER LOT



DOUBLE FRONTING



STANDARD WITHOUT LANE

